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Science Capstone

Mapping Elemental Abundances in the Milky Way Across Galactic Time

1. Introduction

Understanding the chemical history of the Milky Way is critical to constraining models for the development of the Milky Way and other galaxies (Martig et al. 2016). One of the main ways this can be done is through the study of age-metallicity relations. Understanding the age-metallicity relation would allow scientists to know when and where the diverse elements which make up our chemical existence originate, as well as continue to constrain models of nucleosynthesis. Nucleosynthesis is the process by which heavy elements are synthesized by stars, and has been studied, especially in the Sun and its stellar neighborhood, for decades (Burbidge et al. 1957). This study proposes to use the large data-sets of stellar abundances now available to produce a 3D map of stellar abundances as they developed across Galactic time.

In the last five to ten years, a number of large spectrographic surveys of Milky Way stars have provided a massive amount of data for astronomers to analyze. According to Bellardini et al. (2022), the introduction of surveys such as the Galactic Archaeology with Hermes (GALAH), *Gaia*-ESO, the Large Area Multi-Object Fiber Spectroscopic Telescope, and the Apache Point Galactic Evolution Experiment (APOGEE), millions of spectra from hundreds of thousands of stars have been observed, and the numbers of observations will continue to increase.

This study uses the APOGEE and GALAH surveys as the primary data sources for analysis. APOGEE is a survey of stars in the Milky Way remarkable for its high-resolution, infrared spectra of stars with signal-to-noise ratios greater than 100 (Majewski et al., 2017). The APOGEE survey (including APOGEE-2) was gathered over a decade from 2011-2021, as part of the Sloan Digital Sky Survey (SDS) III and IV, which are scientific collaborations to provide large amounts of publicly accessible astronomical data through observations with the Sloan Digital Sky Survey telescope at the Apache Point Observatory in southeast New Mexico (Gunn et al., 2006) and the Irénée du Pont Telescope (Bown and Vaughan, 1973) at Las Campanas Observatory in northern Chile. The completed APOGEE and APOGEE-2 survey, which include 2,659,178 individual spectra, was released in SDS Data Release 17 (Abdurro'uf et al. 2022).

The spectra obtained in the APOGEE surveys have been processed into the elemental abundances present in observed stars, as well as other stellar parameters including the effective temperature (T_{eff}), surface gravity ($\log g$), total metallicity ($[M/H]$), and alpha element abundance ($[\alpha/M]$) (García Pérez et al. 2016). Alpha elements are the elements produced by alpha process of stellar nucleosynthesis, in which alpha particles and carbon produced by helium burning are fused to create the elements up to iron (Burbidge et al. 1957). The spectra have been processed into these parameters by the APOGEE Stellar Parameters and Chemical Abundances Pipeline (ASPCAP, Holtzman et al. in prep.). In basic terms, ASPCAP works by comparing synthetically produced spectra with known parameters to the observed spectra in the APOGEE data to find a χ^2 best-fit between one of these synthetic spectrum models and the observed spectrum (García Pérez et al. 2016).

The GALAH survey also seeks to provide high-spectral resolution datasets for almost one million stars across a diverse sample of Milky Way stars. This survey made its observations from

the High Efficiency and Resolution Multi Element Spectrograph (HERMES) instrument at the 3.9m Anglo-Australian Telescope at Siding Spring Observatory. This instrument is specifically designed to provide high-quality data for studying the chemical history of the Galaxy (Sheinis et al. 2015). This survey provides data for up to 32 different elements using synthetic spectrum fits similar to ASPCAP, complimented with neural network models (Buder et al. 2024).

These stellar abundances, along with the positions of these stars in the Galaxy and estimates of their age, can be used to find relations between stellar abundances and the general times and locations of a star’s formation, a process known as “chemical tagging” (Bellardini et al. 2022). Multiple chemical evolution models suggest that the metallicity of the Galactic Interstellar Medium, the material from which stars form, should increase with time, resulting in a correlation between age and metallicity (Hasselquist et al. 2019). Similarly, the “inside-out” theory of galaxy formation, where the galactic disk forms fastest in the inner disk and slower at increased radial distance formation, predicts a gradient relation of decreasing metallicity with radius (Bird et al. 2013). These theoretical models provide a basis for suspecting the viability of chemical tagging techniques and to age-metallicity relations in particular.

Many studies in the literature have investigated age-metallicity relations but with varying results. Some early results using stars in the solar neighborhood found a significant age-metallicity relationship (Carlberg et al. 1985), but later studies found only a very weak one (Feuillet et al. 2016). However, some studies have found strong age relations with $[\alpha/M]$ (Feuillet et al. 2016; Hasselquist et al. 2019), the ratio of alpha elements to total metallicity, and others have found relations using particular elements, such as beryllium (Pasquini et al. 2007) and carbon-nitrogen ratios (Martig et al. 2016), at least for particular stellar populations. However, none of these make use of a combined dataset as large as this study.

2. Hypothesis and Study Goals

This study will analyze the data from the APOGEE SDSS DR17, GALAH DR4 and related datasets in order to (1) search for or confirm chemical relationships providing information about the age and orbital radius of stars in the Milky Way and (2) use these relations to provide informative visualizations of the chemical landscape of the Milky Way across space and time.

3. Data Preparation (Methodology)

3.1 Dataset Acquisition

APOGEE Data. The APOGEE DR 17 dataset was obtained from the SDSS-IV website, which can be found at the URL https://www.sdss4.org/dr17/irspec/spectro_data/. This study uses the full “allStar” dataset in the allStar-dr17-synspec_rev1.fits file. 23 elemental abundances were extracted from this file. Age data was obtained through DR17’s value added catalogs (VACs). Two separate age datasets were obtained: DistMass (Stone-Martinez et al. 2024) and AstroNN (Mackereth et al. 2019). Both VACs use neural networks to estimate age. DistMass maps select stellar parameters and carbon and nitrogen abundance to mass, which can be used to derive age for evolved giant stars and is trained on asteroseismic and main-sequence isochrone masses. AstroNN maps the complete stellar spectrum to age and is also trained on asteroseismic ages. Distances to the stars in the APOGEE dataset can be found within SDSS’s “allStar” file, but were calculated from the Gaia Space Telescope’s (Gaia Collaboration et al. 2016) Early Data Release 3 (Gaia Collaboration et al. 2021) parallax values according to the statistical methods employed by Bailer-Jones et al. (2021). This study uses the median distance values derived from geometric parallax.

GALAH Data. The GALAH DR4 dataset was obtained from the GALAH survey website, and can be found at the URL

https://cloud.datacentral.org.au/teamdata/GALAH/public/GALAH_DR4/catalogs/. The “allStar” GALAH data file was used. Abundance data for 29 elements were obtained from this file. Age data for GALAH sources is included in the “allStar” file and is calculated “on the fly” using isochrone matching (Buder et al. 2024). Most of the distances used by the GALAH dataset use the values from Bailer-Jones et al. (2021), while some leverage information from star clusters included in Cantat-Gaudin & Anders (2020), Vasiliev & Baumgardt (2021) and Baumgardt & Vasiliev (2021). Based on cross referencing between Gaia source-ids for each source in the APOGEE and GALAH datasets, it was possible to determine that there is an overlap of approximately 3.189%.

3.2 Preparing the Datasets

Preparing APOGEE Dataset. Not all of the 733,901 sources within the APOGEE “allStar” dataset have useful abundance data, distance, and age data. After filtering out stars with no metallicity or distance data, 638,591 sources remained. The sample was further narrowed by filtering out sources based on their ASPCAP quality bitmask flags. Only sources which were not flagged with metallicity warnings, alpha-abundance warnings, or carbon and nitrogen abundance warnings and had a signal-to-noise greater than or equal to 70 were used. This left 488,630 stars. Of these stars, all had AstroNN age data and 356,317 had DistMass age data. The DistMass dataset contains data based on training using three different datasets, each both corrected and uncorrected. This study uses the DistMass figures trained on asteroseismic data from Sharma et al. (2016) because this is the latest sample. All three models are largely similar, differing from each other only in some small population of stars which has poor agreement with AstroNN, as

can be seen in Figure 1. Both datasets contain accuracy estimates. In order to improve accuracy and remove outliers, only the top 50% of each age dataset were used. This left 319,296 stars in both datasets, with an overlap of about 117,576 stars.

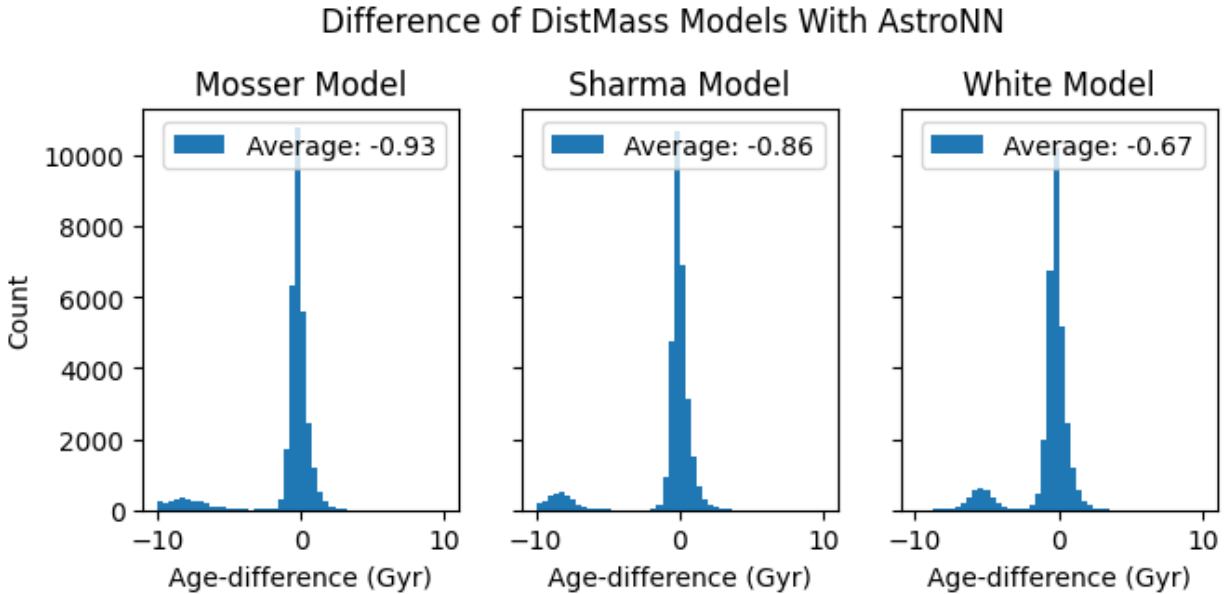


Figure 1. Histograms show the difference in age value for stars in the intersection of the DistMass and AstroNN datasets. Each of the three models shows exceptional agreement with AstroNN, except for a small population with a difference mean far from zero.

Preparing the GALAH Dataset. As with APOGEE, the GALAH dataset was reduced to filter out bad data. In accordance with the recommendations made by the GALAH team (cite), only stars with no stellar parameter warning flags and a per-pixel signal-to-noise ratio greater than 30. From the 917,588 original stars, 625,213 remained after this reduction. When individual abundances were considered, stars with relevant warning flags were removed, though this differs between elements. No official estimate of errors is available for the age values within the GALAH dataset, making it difficult to address outliers. However, all stars with age values greater than 14 Gyr are removed as obviously erroneous.

3.3 Software

This study was made possible by the tools provided in the Astropy python library (Astropy Collaboration et al. 2022), particularly in handling FITS files. Figures were made possible by the Matplotlib python library (Hunter 2007). Efficient numerical operations on large arrays were made possibly by the Numpy python library (Harris et al. 2020). Finally, statistical functions found in the “stats” module of the Scipy python library (Virtanen et al. 2020) were also used.

4. Results

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